

Learning & Reference Links

ML Superset — Smarter Bank Campaigns

Ordered to match the assignment. Work top to bottom. Format: Topic — Resource name — link

0. Dataset

- Dataset & variable definitions — UCI Machine Learning Repository: Bank Marketing — <https://archive.ics.uci.edu/dataset/222/bank>

1. Foundations

- pandas basics (EDA, dtypes, groupby) — pandas User Guide — https://pandas.pydata.org/docs/user_guide/index.html
- How scikit-learn's API works — scikit-learn: Getting Started — https://scikit-learn.org/stable/getting_started.html
- Train/test split & cross-validation (docs) — scikit-learn: Cross-validation guide — https://scikit-learn.org/stable/modules/cross_validation.html
- Cross-validation (video) — StatQuest: Machine Learning Fundamentals — Cross Validation — <https://www.youtube.com/watch?v=fSytzGwwBVw>
- Confusion matrix (video) — StatQuest: The Confusion Matrix — <https://www.youtube.com/watch?v=Kdsp6soqA7o>
- Sensitivity & specificity (video) — StatQuest: Sensitivity and Specificity — <https://www.youtube.com/watch?v=vP06aMoz4v8>

2.1 Preprocessing — Pipeline + ColumnTransformer

- Pipeline class (docs) — scikit-learn: sklearn.pipeline.Pipeline — <https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html>
- ColumnTransformer class (docs) — scikit-learn: sklearn.compose.ColumnTransformer — <https://scikit-learn.org/stable/modules/generated/sklearn.compose.ColumnTransformer.html>
- Pipelines & composite estimators (user guide) — scikit-learn: Pipelines and composite estimators — <https://scikit-learn.org/stable/modules/compose.html>
- Worked example, numeric + categorical together — scikit-learn Example: Column Transformer with Mixed Types — https://scikit-learn.org/stable/auto_examples/compose/plot_column_transformer_mixed_types.html
- Pipeline/ColumnTransformer walkthrough (video) — Column Transformer in Machine Learning | How to use Sklearn ColumnTransformer — <https://www.youtube.com/watch?v=IPYaxjBiKGE>

2.2 Baseline — Logistic Regression

- Logistic Regression (video) — StatQuest: Logistic Regression — <https://www.youtube.com/watch?v=yYKR4sgzI8>
- LogisticRegression class (docs) — scikit-learn: sklearn.linear_model.LogisticRegression — https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

2.3 Bagging — Random Forest

- Random Forests, Part 1 (video) — StatQuest: Random Forests Part 1 – Building, Using and Evaluating — https://www.youtube.com/watch?v=J4Wdy0Wc_xQ
- Random Forests, Part 2 — missing data (video) — StatQuest: Random Forests Part 2 – Missing data and clustering — <https://www.youtube.com/watch?v=sQ870aTKqiM>
- RandomForestClassifier class (docs) — scikit-learn: sklearn.ensemble.RandomForestClassifier — <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>
- Ensemble methods (user guide) — scikit-learn: Ensemble methods — <https://scikit-learn.org/stable/modules/ensemble.html>

2.4 Boosting — AdaBoost / Gradient Boost / XGBoost

- AdaBoost (video) — StatQuest: AdaBoost, Clearly Explained — <https://www.youtube.com/watch?v=LsK-xG1cLYA>
- Gradient Boost, Part 1 — regression main ideas (video) — StatQuest: Gradient Boost Part 1 – Regression Main Ideas — <https://www.youtube.com/watch?v=3CC4N4z3GJc>
- Gradient Boost, Part 2 — regression details (video) — StatQuest: Gradient Boost Part 2 – Regression Details — <https://www.youtube.com/watch?v=2xudPOBz-vs>
- Gradient Boost, Part 3 — classification (video) — StatQuest: Gradient Boost Part 3 – Classification — <https://www.youtube.com/watch?v=jxuNLH5dXCs>
- Gradient Boost, Part 4 — classification details (video) — StatQuest: Gradient Boost Part 4 – Classification Details — <https://www.youtube.com/watch?v=StWY5QWMXCw>
- XGBoost, Part 1 — regression trees (video) — StatQuest: XGBoost Part 1 – XGBoost Trees for Regression — <https://www.youtube.com/watch?v=OtD8wVaFm6E>
- XGBoost, Part 2 — classification trees (video) — StatQuest: XGBoost Part 2 – XGBoost Trees for Classification — <https://www.youtube.com/watch?v=8b1JEDvenQU>
- GradientBoostingClassifier class (docs) — scikit-learn: sklearn.ensemble.GradientBoostingClassifier — <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.GradientBoostingClassifier.html>
- XGBoost library (docs) — XGBoost Documentation — <https://xgboost.readthedocs.io/en/stable/>
- LightGBM library (docs) — LightGBM Documentation — <https://lightgbm.readthedocs.io/en/latest/>

2.5 LDA & QDA

- Linear Discriminant Analysis (video) — StatQuest: Linear Discriminant Analysis (LDA), Clearly Explained — <https://www.youtube.com/watch?v=azXCzI57Yfc>
- LDA/QDA (user guide) — scikit-learn: Linear and Quadratic Discriminant Analysis — https://scikit-learn.org/stable/modules/lda_qda.html
- LinearDiscriminantAnalysis class (docs) — scikit-learn: sklearn.discriminant_analysis.LinearDiscriminantAnalysis — https://scikit-learn.org/stable/modules/generated/sklearn.discriminant_analysis.LinearDiscriminantAnalysis.html
- QuadraticDiscriminantAnalysis class (docs) — scikit-learn: sklearn.discriminant_analysis.QuadraticDiscriminantAnalysis — https://scikit-learn.org/stable/modules/generated/sklearn.discriminant_analysis.QuadraticDiscriminantAnalysis.html

2.6 Class imbalance

- SMOTE concept (video) — SMOTE Oversampling Algorithm for Class Imbalance, Explained — <https://www.youtube.com/watch?v=v-fW2ZdjNCY>
- imbalanced-learn library (docs) — imbalanced-learn Documentation — <https://imbalanced-learn.org/stable/>
- imbalanced-learn source — GitHub: scikit-learn-contrib/imbalanced-learn — <https://github.com/scikit-learn-contrib/imbalanced-learn>
- SMOTE-safe pipeline class (docs) — imbalanced-learn: imblearn.pipeline.Pipeline — <https://imbalanced-learn.org/stable/references/generated/imblearn.pipeline.Pipeline.html>
- Original SMOTE paper — Chawla et al., "SMOTE: Synthetic Minority Over-sampling Technique," JAIR (2002) — <https://www.jair.org/index.php/jair/article/view/10302>

2.7 Time-based split

- TimeSeriesSplit class (docs) — scikit-learn: sklearn.model_selection.TimeSeriesSplit — https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.TimeSeriesSplit.html
- Time-series cross-validation (user guide section) — scikit-learn: Cross-validation — Time Series Split — https://scikit-learn.org/stable/modules/cross_validation.html#time-series-split

2.8 Evaluation — metrics, precision@k, lift@k

- ROC and AUC (video) — StatQuest: ROC and AUC, Clearly Explained! — <https://www.youtube.com/watch?v=4jRBRDbJemM>
- All classification metrics in one place (user guide) — scikit-learn: Model evaluation — https://scikit-learn.org/stable/modules/model_evaluation.html
- Precision-recall curve function (docs) — scikit-learn: sklearn.metrics.precision_recall_curve — https://scikit-learn.org/stable/modules/generated/sklearn.metrics.precision_recall_curve.html
- Average precision / PR-AUC function (docs) — scikit-learn: sklearn.metrics.average_precision_score — https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html

3.1 Clustering fundamentals

- K-means clustering (video) — StatQuest: K-means clustering — <https://www.youtube.com/watch?v=4b5d3muPQmA>
- Hierarchical clustering (video) — StatQuest: Hierarchical Clustering — <https://www.youtube.com/watch?v=7xHsRkOdVwo>
- All clustering methods in one place (user guide) — scikit-learn: Clustering — <https://scikit-learn.org/stable/modules/clustering.html>

3.2 Feature scaling for clustering

- StandardScaler class (docs) — scikit-learn: sklearn.preprocessing.StandardScaler — <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

3.3 K-means / Agglomerative / DBSCAN

- KMeans class (docs) — scikit-learn: `sklearn.cluster.KMeans` — <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>
- AgglomerativeClustering class (docs) — scikit-learn: `sklearn.cluster.AgglomerativeClustering` — <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.AgglomerativeClustering.html>
- DBSCAN class (docs) — scikit-learn: `sklearn.cluster.DBSCAN` — <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html>
- DBSCAN concept (video) — StatQuest: Clustering with DBSCAN, Clearly Explained — <https://www.youtube.com/watch?v=RDZUdRSDOOK>
- Visual comparison of methods (example) — scikit-learn Example: Comparing different clustering algorithms on toy datasets — https://scikit-learn.org/stable/auto_examples/cluster/plot_cluster_comparison.html

3.4 Choosing k / validating clusters

- Silhouette score concept (video) — What Is Silhouette Score? Visually Explained — <https://www.youtube.com/watch?v=mWdxGajUiS8>
- `silhouette_score` function (docs) — scikit-learn: `sklearn.metrics.silhouette_score` — https://scikit-learn.org/stable/modules/generated/sklearn.metrics.silhouette_score.html
- Silhouette analysis walkthrough (example) — scikit-learn Example: Selecting the number of clusters with silhouette analysis — https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

3.5 PCA visualization

- PCA fundamentals (video) — StatQuest: Principal Component Analysis (PCA), Step-by-Step — <https://www.youtube.com/watch?v=FgakZw6K1QQ>
- PCA implemented in Python (video) — StatQuest: PCA in Python — <https://www.youtube.com/watch?v=Lsue2gEM9D0>
- PCA class (docs) — scikit-learn: `sklearn.decomposition.PCA` — <https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>
- Decomposition / PCA (user guide) — scikit-learn: Decomposing signals — PCA — <https://scikit-learn.org/stable/modules/decomposition.html#pca>

4. Final submission

- Reproducibility & `random_state` (docs) — scikit-learn: Common pitfalls — Controlling randomness — https://scikit-learn.org/stable/common_pitfalls.html#randomness
- Project structure template (optional) — GitHub: `drivendataorg/cookiecutter-data-science` — <https://github.com/drivendataorg/cookiecutter-data-science>